

Board exams, rote learning and the learning crisis

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THIS article starts by explaining why the author believes there is a learning crisis in India, how it probably originated and why it has persisted despite resources expended over many years. It goes on to discuss rote learning, seen as one of the root causes for the learning crisis, and board exams, which sustain rote learning. Finally, it talks about some proposed solutions.

Multiple studies have shown that there is a clear gap between what we expect children to know and understand, and what they are able to do. The reason we consider it a crisis in India (rather than just a learning gap) is that both in relative and absolute terms, the situation is extremely serious. India stood 73 out of 74 countries in the 2009-10 PISA test. Meanwhile, the ASER tests show repeatedly that about 50% of grade 5 children cannot read a grade 2 text. The results in arithmetic are equally poor.

We live in a world where basic education and critical thinking are becoming prerequisites for most jobs. If a significant percentage of Indians do not acquire basic skills, this represents a danger not just to progress, but even peace and stability in the region and beyond.

What caused the learning crisis and why haven't years of effort and resources addressed it? To understand this, we shall introduce the ideas of *technical and administrative paradigms*.¹ Every sector needs solutions in both the technical space and administrative (or governance) space. To take telecom as an example, the various core technologies (like 4G and 5G) relate to the technical paradigm. Practices like the process to allot spectrum correspond to the governance or administrative paradigm. A sector can flourish only if there is focus and progress on both these paradigms in that sector.

In public education, the focus seems to have been disproportionately on the administrative paradigm. The technical paradigm here refers to all aspects of curriculum, teaching techniques, research into learning and assessments, while the administrative paradigm covers areas like teacher accountability, teacher-student ratios, delivering textbooks on time, conducting examinations, and so on. There is an unstated assumption that the technical dimension is easy or trivial. This is seen both in comments like: 'What's there to teaching class 3 children – anyone should be able to do it...' and in administrative duties crowding out educational ones for many people working in the system.

We can now examine what sowed the seed for the current learning crisis. Our current education system was actually designed in the mid-1800s with a goal of producing effective workers and intermediaries to support the British Empire. While the world around has changed significantly and at an increasingly faster pace, education systems and curriculums tend to display a lot of inertia. The few changes that have been made (moving to the 10+2 system, semesterization, etc.) have tended to be based more on the administrative rather than the technical paradigm. The disconnect in the vision, in the language used and the lack of change over time have tended to make the enterprise of education/teaching more mechanical alongside a disproportionate focus on making it 'easier to test'.

This has led to what is called rote learning. We shall define and discuss it in depth later. Seen through the lens of the administrative paradigm, education is about children attending classes in which teachers are present, those children taking tests from time to time, and examinations at the end of the year. The very concept of

‘learning’ (as opposed to marks in a test) needs an appreciation of the technical paradigm.

Teacher training, in the administrative paradigm, is measured in terms of number of teachers trained and hours of training. This kind of training is ineffectual when the students are not from middle class backgrounds with educated parents and access to learning material. As the system expanded, a large number of teachers had to be recruited, but with training focused more on numbers, teacher quality suffered further. Today, when evidence of poor learning can no longer be ignored, it is teachers who are blamed for their poor skills. Of course, teachers have poor skills, but that’s the symptom, not the cause of the problem.

How does teacher capacity building differ when seen from the lens of the technical paradigm? The focus would be on what training programmes are being offered, their impact if any, and correlating the actual learning gaps with those teacher training programmes. In many states, it has been observed that Block and Cluster Resource Coordinators do not play an academic support role (the original intention) but end up with administrative responsibilities – a classic example of the administrative crowding out the technical.

Let us now turn to the issue of rote learning and its role in sustaining the learning crisis. Reforming an education system is not easy, especially when the problems have accumulated over decades. However, there is one key lever that is available in tackling the learning crisis that we face today, and that is addressing rote learning.

What is rote learning? Many people think that memorizing or learning things by heart is rote learning. Rather, rote learning is learning something meaningless or which the learner does not understand, only or primarily because it is ‘required’ to be learnt for a test or because the teacher expects it. If one memorizes multiplication tables (and understands what ‘5 7’s are 35’ means), it is not rote learning and actually improves your arithmetic fluency. But if one learns a formula or a definition by heart without quite understanding it, simply because it will be asked in the test, that is rote learning. If a teacher asks students to name three things made of wood, and then expects them to name the same three objects mentioned in the textbook, that is rote learning. If students can recite the definition of a triangle, but cannot identify a narrow three-sided figure as a triangle or if they memorize an answer in English literature or History containing terms they don’t understand, those are examples of rote learning.

People sometimes argue that though bad, rote learning is not as harmful as made out. We disagree. Once rote learning starts dominating a system, it kills meaning and purpose, not just for students but even for teachers. Learning can be a most exciting and passionate endeavour. Rote learning kills that curiosity and passion. What begins as rote learning in a chapter in Mathematics or Language becomes a system of accepting what a textbook or a teacher says, even if it does not make sense. It discourages thinking and encourages accepting things either because others accept it or because ‘that is the way things are always done’. This expands into a rote system where marks, certificates and jobs are valued, and not real life skills. This is the perfect formula for dividing things between ‘what happens in textbooks and the school’ and ‘what happens in real life’, a distinction that is very clear in today’s world (for example, ‘speak in mother tongue at home but in English in school’).

The opposite of rote learning is ‘learning with understanding’. Sometimes people mistakenly believe that learning with understanding is merely a higher level of learning beyond rote learning. But students cannot attain rote learning first and then ‘learning with understanding’ because very different skills are needed and developed in each of them.

Many countries complain about rote learning in their educational system. However, its extent and intensity are extremely high in India. For example, consider a common question in primary school Mathematics – calculating the least common multiplier (or LCM) of two numbers like 12 and 20. In many school systems, children follow a procedure to reach the answer of 120. It is also likely that in many of these situations, children do not fully understand the significance of what they are doing. The difference we have found is that in India (and some other countries), teachers accept this and say, ‘This is okay, we all learnt it that way only’. But in other countries, less tolerant of rote, teachers acknowledge that students often do not understand without implying that it is okay.

In 2006 and 2010, our company, jointly with Wipro, conducted studies which were published in *India Today*,² *The Hindu* and *Mint* publications. These studies showed that the problem of rote learning affects even our ‘top’ schools, and the performance of grade 4 students in these top schools was below international average. Further, the situation did not improve in the intervening years.

The cause and persistence of rote learning can be ascribed to one key attribute – our board exams, which influence all our school exams. On the positive side, this means that a lot can be achieved by reforming the pattern of questions in our board exams.

People are sometimes surprised to know that board exams can make learning more rote-based in the primary classes. We observed this phenomenon when we started a school, initially only up to class 5 and then added one higher class every year. In the initial years we found the teachers and classes focused on learning with understanding. But once teachers had to prepare their students to write a board exam, there was a marked shift which had a clear downstream effect. Teachers of class 9 would expect children to be prepared in a certain way in class 8 to be able to answer board questions well. These included things like ‘not writing the answer in one’s own words’, ‘using simple language’, ‘using terms and expressions from the textbook where possible’, ‘using examples given from the textbook’, ‘sticking to the information in the textbook even if it was wrong or outdated’ – perfect recipes for rote learning. This was the exact opposite of what we would normally tell teachers and students, and this is how rote learning in the board exam (which is the ‘goalpost’), affects the system as a whole.

There are serious problems at every stage of the board exam and all contribute to the learning crisis: the very validity of questions in the board exams – the NCF 2005 included position papers on many different topics. Here is an excerpt from the paper on examination reforms 2004-05:³

‘The core of the exam system is the exam paper. This may seem almost a tautological assertion but, given the lack of attention paid by most boards to the quality of the actual exam paper, it is necessary to make it. The question papers remain seriously problematic.’

Question paper sets from the most recent (March 2004) 10th and 12th grade exams were collected for detailed study. Attention was focused on paper sets from five boards popularly perceived to be the best in the country. The exercise was an eye-opener:

* *What is the weight of the pituitary gland?* (non-essential, the gland should be studied for its crucial function, perhaps even for its structure, but hardly its weight!)

* *Who were the parents of Benito Mussolini?* (Irrelevant.)

* *How many members are there in the U.N.O.?* (Transient.)

* *Who was called Modern Messiah?* (The term was employed by the textbook writer to describe Karl Marx but has no wide currency outside the textbook.)

* *Describe the method of irrigation prevalent in India* (Takes as a given fact that there is only one such method in India, perhaps because the textbook has mentioned only one!)

* *Our highest import is from (Hong Kong, Italy, Kuwait)?* (No correct answer provided – at least for any year in the last quarter century.)

Unfortunately, those criticisms of 2005 are equally valid today.

The problem of poor and invalid questions does not plague the board exams alone. As part of an NCTE committee examining the reasons for only 1% of candidates passing the Central Teacher Eligibility Test (CTET), I had the opportunity to analyse questions from the November 2012 CTET test along with the performance data. Note that CTET is set by the CBSE (and is considered a much better test than the state TETs). Here are two questions from the paper.

a) One Pashmina shawl is as warm as normal sweaters andhours are taken to weave a plain Pashmina shawl:

(i) 10; 200; (ii) 06; 250; (iii) 06; 200; (iv) 10; 250

b) In order to instil a positive environment in a primary class, a teacher should:

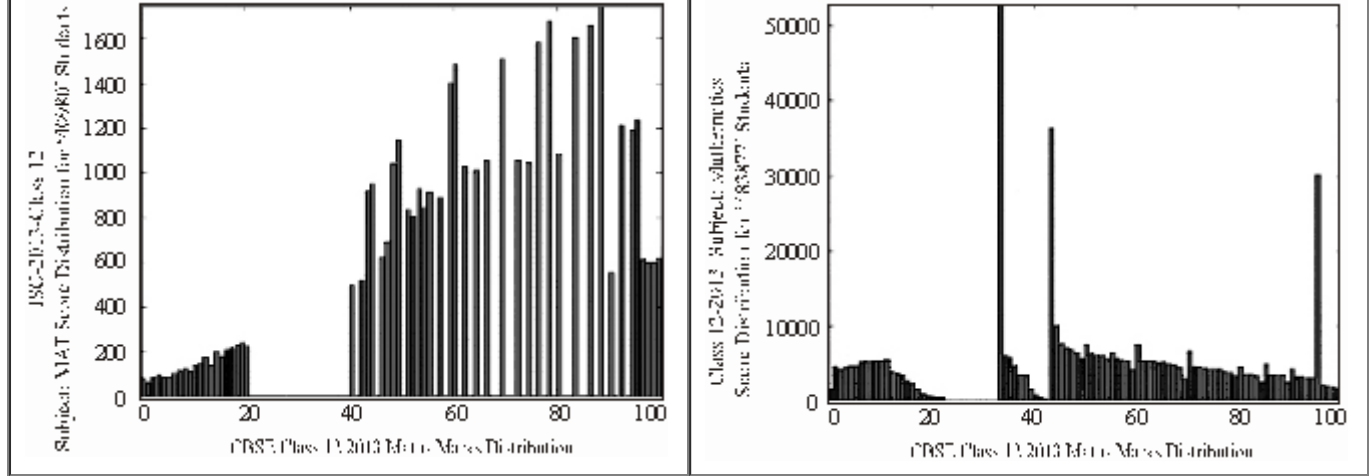
(i) wish each child in the morning; (ii) not discriminate and set the same goal for every child; (iii) allow them to make groups on their own on the basis of sociometry during group activities; (iv) narrate stories with positive endings.

The basis of the data for the first question (apart from how comparative warmth is measured) is unclear. In the second question, the correct answer was marked as (1) and experts commented that most of the other options seem better ones. Let me add that I have not cherry-picked questions, a majority of questions had issues with validity or ambiguity or level of difficulty.

Somehow this issue of quality and appropriateness of questions in exams has never got much attention, nationally or internationally. Internationally, this is not seen as a big problem in the developed world and researchers there probably assume that questions would be fine. However, recently a working paper by Newman Burdett compared board exam papers from India, Pakistan, Uganda, Nigeria and a Canadian province, Alberta and found that the extent of rote questioning in India and Pakistan was even higher than the African countries: ‘In India and Pakistan, higher-order skills were almost entirely lacking and the focus was very much on recall of very specific rote-learned knowledge... In the two African countries, this rote-learning approach seemed less extreme.’⁴

The process to set and check the papers and keep them error-free: The process of making papers has remained a largely manual process where question makers submit papers in sealed covers and a paper is chosen at random for the actual test. These days, it is possible to use software to have questions solved and checked independently so that errors and even ambiguities can be all but eliminated.

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The way board exams are scored. The graphs below from the work of Prashant Bhattacharji⁵ show how the marks of ISC (left) and CBSE (right) students in the Maths class 12 exam in 2013 are distributed. This is data for every student who took the tests.

There are two points to note – each of which represents a serious problem if not a scam. (i) These are not normal curves, which is very odd and would need explanation. Notably all spikes at marks like 36, 43 and around 96 in CBSE and 88, 95 and other scores in ISC are inexplicable and unfair. (ii) Both graphs show gaps, meaning no student scored those marks. This is not possible under normal conditions and suggests grace marks, also unfair to other students.

Why is this happening? It is said, ‘Never attribute to malice that which is adequately explained by stupidity.’ We see this more as a lack of capacity in the boards rather than a desire to manipulate. Yet, this is extremely unfair for a high stakes school leaving exam. It must be noted that nothing much changed based on these revelations in 2014; the author detected the same irregularities the next year.

There is also the issue of comparability of board marks. One can understand that marks of two different boards are not comparable. But for many years, even marks across CBSE zones were not comparable (students score higher in the Chennai region of CBSE compared to the Delhi region). These are apparently done to allow the students to compete with the local state boards but are unfair for students who seek admission in a different region.

Earlier, we have discussed in some depth why we feel there is a learning crisis, its original causes and why it persists despite so much expenditure of effort and resources. We have also discussed the key role of rote learning and many board exam related problems. In this section, we shall discuss some solutions.

The first principle that we should remember is that deep problems usually do not have quick fixes. All the solutions proposed will take at least three to five years to have an impact. These solutions focus on fundamentally addressing key issues like rote learning and low capacity in the system.

Focus on foundational learning: For any student, learning is like constructing a building – there is a foundation and then there are stages with each serving as the base for the next. The foundation for all learning is that *by the age of 8 or 9*, children must be able to read fluently, do arithmetic operations and develop basic critical thinking skills. The age deadline (corresponding to class 3 or 4) is critical. Either by law or through a campaign (like Swachh Bharat), this should be set as a common goal to be achieved by states and all types of schools in the next five years

(with intermediate goals). A number of enablers in the form of resources for teachers, books and apps for students and information for parents and society are needed to make this happen. This will include research and assessments. But this will be the first (and most important) step in the battle to defuse the learning crisis.

Build systemic and institutional capacity: Capacity building is often used to refer to training of teachers and educational personnel. However, this ‘people capacity building’ has to be preceded by ‘institutional capacity building’. This essentially means that key educational institutions – research institutions like NCERT, teacher training institutions like colleges of education and DIETs, assessment institutions like the boards and others like the NCTE – must be strong and should continuously strengthen their expertise in their core functional area. They should evoke respect from the academic as well as the practitioner community. The steps for this are appropriate leadership and hiring, research as a key role and providing autonomy to institutions. Do we, for instance, see NCERT in the same light as an IIT or IIM? Is that not possible to imagine and make happen?

Systemic capacity is the ‘body of knowledge’ that gets created and advances a sector. For example, how does one get children in very low income families to become effective readers? How does one create high quality computer based assessments in the Indian context? While some of this knowledge can and should draw upon what is already known internationally, there will be a lot of knowledge that will have to be created (and hopefully contributed to the international community). If Google can develop techniques of working with low bandwidth connections and offline maps in India (and then spread that to the rest of the world), can we not discover educational approaches that work in our conditions and use that to advance the field internationally?

Build a new Science of Learning: The Science of Learning is a new interdisciplinary science which draws upon school subjects like mathematics, science and language as well as pedagogy, psychology, cognitive sciences, neuroscience, AI and related fields to answer the question, ‘How can human beings, especially children, learn better?’ Just like medical science studies disease, symptoms, diagnostic techniques and treatments and builds on data to provide answers that doctors can use to treat patients, the Science of Learning uses data on learning, techniques for learning and teaching insights to provide insights to teachers on a regular basis. It allows teachers to use the art of teaching they practice and draw upon the Science of Learning whenever they need, to lead to improved student learning. Some Science of Learning researchers may study ways to improve reading skills in children, while others may research issues with Algebra learning in children, while still others may focus on effectiveness of AI based apps for students.

There is no doubt that we are in the midst of a learning crisis today. But we also have rich human resources that we can focus to not just solve this crisis for ourselves, but come up with techniques and solutions by strategically focusing on this area. Every country needs these solutions and there are, as yet, no ready answers. Can we convert this problem into an opportunity?

Footnotes:

1. Based on researches by Mariana Mazzucato including https://www.ted.com/talks/mariana_mazzucato_government_investor_risk_taker_innovator

2. <https://www.ei-india.com/wp-content/uploads/2017/02/India-Today-printable-November-27-2006-Whats-Wrong-With-Our-Teaching.pdf>

3. [http://www.ncert.nic.in/html/pdf/school curriculum/position_papers/examination_reforms.pdf](http://www.ncert.nic.in/html/pdf/school_curriculum/position_papers/examination_reforms.pdf)
4. https://www.riseprogramme.org/sites/www.riseprogramme.org/files/publications/RISE_WP-018_Burdett.pdf
5. <http://www.thelearningpoint.net/home/examination-results-2013/exposing-cbse-and-icse>

